

DOCUMENTATION FOR THE RATPOINTS PROGRAM

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1 Introduction

This paper describes the `ratpoints` program. This program tries to find all rational points within a given height bound on a hyperelliptic curve in the most efficient way possible.

2 History and Acknowledgments

This program goes back to an implementation of the ‘quadratic sieving’ idea by **Noam Elkies** that was around in the early 1990s. My own first contribution was to replace the char arrays that were used to store the sieving information by bit arrays, in 1995. **Colin Stahlke** then made use of the `gmp` library, so that points could be checked exactly, and implemented the selection of sieving primes according to their likely success rate, in 1998. After that, I successively put in numerous improvements (and some bug fixes). For details of how the program works, see Section 8 below.

Along with **Noam Elkies** and **Colin Stahlke**, I would like to thank **John Cremona** and **Sophie Labour** for bug reports and suggestions for improvements. Further thanks are due to **Bill Allombert** for a first implementation of the use of 256-bit (and possibly 512-bit) registers.

3 Availability

The `ratpoints-2.2.4` package can be downloaded from my homepage, see [rat], or from github at <https://github.com/MichaelStollBayreuth/ratpoints>.

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4 Installation

This section describes the installation procedure under Linux.

4.1 Extract the archive

```
> tar xzf ratpoints-2.2.4.tar.gz
```

This sets up a directory `ratpoints-2.2.4` containing the various files that belong to the installation.

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4.2 Build the program and library

Do

```
> cd ratpoints-2.2.4
> make all
```

This will build the library `libratpoints.a`, the executable `ratpoints`, and also run `pdflatex` to generate this documentation.

By default, the program will use primes up to 127. If you intend to do computations with large height bounds or with curves that you expect to have many points, it may make sense to increase the range of primes. This can be achieved via

```
> make all PRIME_SIZE=8
```

perhaps after a

```
> make distclean
```

to remove the files that were generated previously. The `PRIME_SIZE` argument can be given any value from 5 to 10; otherwise it is taken to be 7. The precise meaning is that the program will work with primes $< 2^s$, when `PRIME_SIZE = s`.

The program now uses SSE instructions if available, working with 128-bit registers. Compared to using word-size registers, this results in a noticeable speed-up. If you don't want this, do

```
> make distclean
> make all CCFLAGS=-UUSE_SSE
```

As of version 2.2, `ratpoints` can also use 256-bit AVX registers. To activate this, change the line `CCFLAGS1 = ${CCFLAGS128}`

in `Makefile` into

```
CCFLAGS1 = ${CCFLAGS256}
```

before building `ratpoints`. This usually gives a further speed-up. You may have to change `-mavx2` in the line

```
CCFLAGS256 = -DUSE_AVX -mavx2
```

into `-mavx` or leave it out altogether, depending on the capabilities of your CPU. You will notice that AVX2 (or AVX) is not supported, when the executable throws an 'unknown instruction' error or similar.

There is also a variant using 512-bit registers, activated by

```
CCFLAGS1 = ${CCFLAGS512}
```

which requires a CPU with AVX512F capability. Note that this variant has so far only been tested indirectly (see the change log entry below), since I have no such CPU available; you should therefore run `make test` before relying on it. It is also worth timing the 256-bit and the 512-bit version against each other on your machine: the sieving loop is limited by memory bandwidth rather than by arithmetic, so wider registers do not automatically pay off, and some Intel server processors reduce their clock frequency when 512-bit instructions are used. The one data point available so far is encouraging, though: on an AMD Ryzen 9 9950X3D (Zen 5 architecture), **Drew Sutherland** measured a speed-up of 13 to 14% over the 256-bit version, for a curve with many rational points.

If you leave out `-mavx512f` and compile with just `-DUSE_AVX512`, then `gcc` will express the 64-byte vectors through narrower operations, so that the code runs on any machine. This is of no use in production (it is slower than the 256-bit version), but it does allow the 512-bit code path to be tested without the corresponding hardware. `gcc` will warn that passing a 512-bit vector argument without AVX512F enabled changes the ABI; this is harmless, as the whole library is built with the same flags.

4.3 Run a test

Run

```
> make test1
```

in the working directory. This will build an executable `rptest` and then run (and time) it. Finally, the output (which was written to a file `rptest.out`) is compared against `testbase`, which contains the output of a sample run. The two should be identical; otherwise the message `Test failed!` will be displayed (on its own on a line).

```
> make test1once
```

runs the same curves again, but with the fields of the argument structure set once before the loop over the curves instead of once per curve, as a program using the library may do (Section 6); `rptest` prints a line whenever the library has changed a field it must leave alone, and the output is compared against `testbase` as before.

```
> make test2
```

calls `ratpoints` on a curve with lots of rational points and with a fairly large height bound, times it, and compares the output to what is expected. This may take a few minutes, so be patient!

```
> make test3
```

runs the invocations of `ratpoints` collected in `test3.sh`, one or more for each of the bugs fixed in version 2.2.4, and compares the output against `testbase3`.

```
> make timing
```

times `ratpoints` on some curve with height bound 400 000. This maybe useful for comparisons. Finally,

```
> make test
```

does all of the above.

4.4 Install

If you like, you can install the library, executable and header file on your system.

```
> sudo make install
```

The executable is copied to `/usr/local/bin/`, the library to `/usr/local/lib/`, and the header file to `/usr/local/include/`. You can change the `/usr/local` prefix via

```
> make install INSTALL_DIR=...
```

4.5 Debugging

In case you found a bug and would like to find out where it comes from, or if you just want to see exactly what the program is doing, you can do

```
> make debug
```

in the working directory. This will build an executable `ratpoints-debug`, which, when run, will dump loads of output on the screen, so it is best to send the output to a file, which you can then study at leisure.

4.6 Cleaning up

In order to get rid of the temporary files, do

```
> make clean
```

To remove everything except the files from the archive, use

```
> make distclean
```

4.7 Requirements

You need a C compiler (like `gcc`, which is the default specified for the `CC` variable in `Makefile`; if necessary, it can be changed there).

In addition to the standard libraries, the program also requires the `gmp` (GNU multi-precision) library [[gmp](#)].

4.8 List of files

The archive contains the following files.

- `Makefile`
- `ratpoints.h` — the header file for programs using `ratpoints`
- `rp-private.h`, `primes.h` — header files used internally
- `gen_find_points.h.c`, `gen_init_sieve.h.c` — short programs that write additional header files depending on the system configuration
- `sift.c`, `init.c`, `sturm.c`, `find_points.c` — the source code for the `ratpoints` library
- `main.c`, `rpctest.c` — the source code for the `ratpoints` and `rpctest` executables, respectively
- `testdata.h`, `testbase`, `testbase2`, `testbase3` — data for the test runs
- `test3.sh` — the script that make `test3` runs; it has to be executable
- `ratpoints-doc-2.2.pdf` — this documentation file
- `gpl-2.0.txt` — the GNU license that applies to this program.

5 How to use `ratpoints`

5.1 Basic operation

Let

$$C : y^2 = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$$

be your curve, and let H be the bound for the denominator and absolute value of the numerator of the x -coordinate of the points you want to find. The command to search for the points is then

```
> ratpoints 'a_0 a_1 ... a_{n-1} a_n' H
```

The first argument to `ratpoints` is the list of the coefficients, which have to be integers, are separated by spaces, and are listed starting with the constant term. There is no bound on the size of the coefficients; they are read as multi-precision integers.

The degree n of the polynomial is limited to `RATPOINTS_MAX_DEGREE` (defined in `ratpoints.h`), which is set to 100 by default.

The program will give an error message when the polynomial (considered as a binary form of the smallest even degree $\geq n$) is not squarefree (e.g., if you specify two leading zero coefficients).

The following subsections discuss how to modify the standard behavior. This is achieved by adding options after the two required arguments. Some of the options have arguments, others don't; the ordering of the options and option-argument pairs is arbitrary (except for options with opposite effect, where the last one given counts, and for the specification of the search intervals, which must be in order).

5.2 Early abort

By default, `ratpoints` will search the whole region that you specify and print all the points it finds. If you just want to find *one* point (for example when dealing with 2-covering curves of elliptic curves), you can tell the program to quit after it has found one point via the `-1` option, e.g.,

```
> ratpoints '-18 116 48 -12 30' 60000000 -1
```

If you don't want to see points at infinity, specify the `-i` option. The two options can be combined: `-i -1` will stop the program after it has found one finite point.

5.3 Changing the output

By default, `ratpoints` prints some general information before and after the points, which are given in the form $(x : y : z)$, one per line. Here, $(x : y : z)$ are the coordinates of the point considered as a point in the $(1, \lceil n/2 \rceil, 1)$ -weighted projective plane, which is the natural ambient space for the curve. The coordinates are integers with x and z coprime and $z > 0$, or $z = 0$ and $x = 1$ (for points at infinity).

There are various ways to change this behavior.

- To suppress all output except the points, use `-q` (for *quiet*).
- To add some more output explaining what the program is doing, use `-v` (for *verbose*). This has no effect if the `-q` option is also given.
- To suppress printing of the points, use `-z`.
- If you only want to list the x -coordinates of point pairs rather than individual points, use `-y`.
- There are four options that influence the format in which the points are printed:

`-f format -fs string-before -fm string-between -fe string-after`

The arguments to all of them are strings; `"\n"`, `"\t"`, `"\""` and `"%"` are recognized and do what you expect. The *format* string can contain markers `%x`, `%y`, `%z` that will be replaced with the x , y , and z -coordinate of the point, respectively. The defaults are empty for *string-before*, *string-between* and *string-after*, and `"(%x : %y : %z)\n"` for *format* when `-y` is not specified; otherwise `"(%x : %z)\n"`.

The effect is as follows. Before any point is printed, *string-before* is output. Then every point is printed according to *format*. Between any two points, *string-between* is output, and after the last point, *string-after* is output.

As an example, consider the effect of

```
-f "[%x,%y,%z]" -fs "{" -fm "," -fe "}\n"
```

5.4 Restricting the search domain

There are two ways to restrict the domain of the search. The first is to restrict the range of denominators considered. This is done via

```
-dl  $d_{\min}$  -du  $d_{\max}$ .
```

For example, to only look for integral points, you can say `-du 1`. By default, the lower limit is 1 and the upper limit is H .

The other way is to restrict the search to a union of (closed) intervals. If you only want to search for points in

$$[l_1, u_1] \cup [l_2, u_2] \cup \dots \cup [l_k, u_k],$$

you can specify this in the form

```
-l  $l_1$  -u  $u_1$  -l  $l_2$  -u  $u_2$  ... -l  $l_k$  -u  $u_k$ .
```

The first `-l` option is optional; l_1 defaults to $-\infty$. Similarly the last `-u` option is optional, and u_k defaults to $+\infty$. The number k of intervals is bounded by `RATPOINTS_MAX_DEGREE` (usually, 100).

5.5 Setting the parameters for the sieve

It is possible to change the number of primes that are used in the various stages of the algorithm. This is done by the following options.

```
-p M -N N -n n
```

This sets the number of (odd) primes that are considered for the sieve to M , the number of primes that are actually used for the sieve to N , and the number of primes used in the first sieving stage to n . The program will, if necessary, reduce M , N , and n (in this order) to ensure that $n \leq N \leq M \leq \text{RATPOINTS_NUM_PRIMES}$. The latter is usually 30 (the number of odd primes $< 2^7$), but will be 53 (the number of odd primes $< 2^8$) if `PRIME_SIZE` is set to 8, etc.

There are two more parameters that can be set.

```
-F D
```

sets the maximal number of ‘forbidden divisors’. If the degree is even and the leading coefficient is not a square, then the denominator of any x -coordinate of a rational point cannot be divisible by a prime p such that the leading coefficient is a non-square mod p . If the leading coefficient is divisible by p , but not a p -adic square, the denominator cannot be divisible by some power of p . The program constructs a list of such forbidden divisors against which to check the denominators, and this option specifies how many of these should be used.

The other option is

```
-S S or -s .
```

In the first form, it sets the number of refinement (interval halving) steps in the isolation of the real components to S . This part of the program computes a Sturm sequence for the polynomial and uses it in order to find a union of intervals that contains the intervals of positivity of the polynomial. This is done by a successive subdivision of the interval $]-\infty, +\infty[$. The number S sets the recursion depth. S can be omitted; then it is given a default value. The option `-s` skips the Sturm sequence computation completely. This also has the effect of removing most of the check for squarefreeness.

5.6 Switching off optimizations

The options `-k` and `-j` can be used to prevent the program from reversing the polynomial, which it usually does if this will lead to faster operation (`-k`), and to prevent the program from using

the Jacobi symbol test on the denominators ($-j$). This last test extends the ‘forbidden divisors’ method described in Subsection 5.5 above by computing the Jacobi symbol (l/d) , where l is the leading coefficient and d is the odd and coprime-to- l part of the denominator. If the symbol is -1 , the denominator need not be considered. The use of these options may be questionable, unless you want to see how much performance is gained by using these optimizations.

5.7 Switching off exact testing of points

The option $-x$ will prevent the program from checking the potential points that survive the sieve whether they really give rise to rational points. This implies that the points that are output may not actually be points. It also effectively sets the $-y$ option, since the y -coordinates are not computed.

5.8 Overriding previous options

The options $-I$, $-Y$, $-Z$, $-S$, $-K$, $-J$, $-X$ can be used to cancel the effect of the corresponding lower-case option (and thereby restore the default behavior), when this occurs earlier in the list of options. This may be useful if you want to set a default behavior that is different from what the program does out-of-the-box (e.g., in a shell script), but want the caller to be able to override this change.

6 How to use the library

It is possible to use the `ratpoints` machinery from within your own programs.

The library `libratpoints.a` provides the following functions.

```
long find_points(ratpoints_args*,
                int proc(long, long, const mpz_t, void*, int*),
                void*);

void find_points_init(ratpoints_args*);

long find_points_work(ratpoints_args*,
                    int proc(long, long, const mpz_t, void*, int*),
                    void*);

void find_points_clear(ratpoints_args*);
```

The passing of arguments to these functions is via the `ratpoints_args` structure, which is defined as follows.

```
typedef struct { mpz_t *cof; long degree; long height;
                ratpoints_interval *domain; long num_inter;
                long b_low; long b_high; long sp1; long sp2;
                long array_size;
                long sturm; long num_primes; long max_forbidden;
                unsigned int flags; ...}
    ratpoints_args;
```

The dots at the end stand for additional fields that are used internally and are not of interest here. All the fields shown are inputs. Some of them the program writes as well, and this is the complete list: the coefficient array and degree, which on return describe the polynomial the program worked with; `num_inter` and `domain`, which on return describe the region it searched; the fields

that ask for a default value when negative or non-positive (`sp1`, `sp2`, `num_primes`, `max_forbidden`, `sturm`, `b_low`, `b_high`, `array_size`), into which it writes the value it used; and the flag bits that report on the run. A caller who fills the structure once and then loops over curves must therefore set before every call whatever it wants chosen afresh. (Up to version 2.2.3 the program also wrote the number of forbidden divisors it had found into `max_forbidden`, and its own decision not to reverse the polynomial into `flags`, which broke exactly such loops; it no longer does.)

When calling `find_points`, the `cof` field must point to an array of size `degree + 1` of properly initialized gmp integers; these are the coefficients of the polynomial. The program can alter the values of the integers in this array: it may reverse the polynomial, and it drops a zero constant term on that occasion, in which case `degree` is lowered by one so that `cof` and `degree` still describe the polynomial the program worked with (the `RATPOINTS_REVERSED` flag says when this has happened). Leading zero coefficients are dropped in the same way, lowering `degree`, without a flag. To prevent the reversal, set the `RATPOINTS_NO_REVERSE` bit in `flags`; this may result in a loss of performance, though.

`height` gives the height bound. It is an error for the degree or the height bound to be nonpositive; in this case the function returns the value `RATPOINTS_BAD_ARGS`.

The field `domain` must contain a pointer to an array of `ratpoints_interval` structures of length at least `num_inter` plus `degree`. This array gives (in its first `num_inter` entries) the intervals for the search region. The type `ratpoints_interval` is just

```
typedef struct {double low; double up;} ratpoints_interval;
```

the meaning of this should be clear. In `ratpoints`, this is set via the `-l` and `-u` options. Usually, you do not want to restrict the range of x -coordinates; then you set `num_inter = 0` (but you still have to fill `domain` with a valid pointer to at least `degree` intervals; a null pointer makes the call return `RATPOINTS_BAD_ARGS`, since the program stores the default interval `]-height, height[` there. If `sturm` is negative, one interval is enough.) The program intersects the given intervals with the region where f is positive (unless `sturm` has a negative value, which may lead to a performance loss) and searches the result; on return, `num_inter` and the first `num_inter` entries of the array describe that result, which is why the array has to be longer than the input. So `num_inter` (and `domain`, if you restrict the search) must be set before every call.

`b_low` and `b_high` carry the lower and upper bounds for the denominator; if non-positive, they are set to 1 and the height bound, respectively. In `ratpoints`, these fields are set by the `-dl` and `-du` options.

`sp1` and `sp2` specify the number of primes to be used in the first sieving stage and in both sieving stages together, respectively. If negative, they are set to certain default values. In `ratpoints`, these fields are set by the `-n` and `-N` options. Similar statements are true for `num_primes` (option `-p`), `sturm` (options `-s`, `-S`) and `max_forbidden` (option `-F`). The field `array_size` specifies the maximal size (in bit arrays; a bit array contains 32, 64, 128, 256, ... bits, depending on the configuration) of the array that is used in the first sieving stage. If non-positive, it is set to a default value. The various default values are defined at the beginning of the header file `ratpoints.h`, where also the maximal degree of the polynomial is set.

The `flags` field holds a number of bit flags.

- `RATPOINTS_NO_CHECK` — when set, do not check whether the surviving x -coordinates give rise to rational points (set by the `-x` option to `ratpoints`).
- `RATPOINTS_NO_Y` — only list x -coordinates (in the form $(x : z) \in \mathbb{P}^1(\mathbb{Q})$) instead of actual points (with a y -coordinate); this is set by the `-y` option to `ratpoints`.

- `RATPOINTS_NO_REVERSE` — when set, do not allow reversal of the polynomial (set by the `-k` option to `ratpoints`).
- `RATPOINTS_NO_JACOBI` — when set, prevent the use of the Jacobi symbol test (set by the `-j` option to `ratpoints`).
- `RATPOINTS_VERBOSE` — when set, causes the procedure to print some output on what it is doing (set by the `-v` option to `ratpoints`).

There are some other flags that are used internally. One of them might be of interest:

- `RATPOINTS_REVERSED` — when set after the function call, this indicates that the polynomial has been reversed (and the contents of the `cof` array have been modified).

The main vehicle for passing information back to the caller is the `proc` function argument together with the pointer `info`. This function

```
int proc(long x, long z, const mpz_t y, void *info, int *quit)
```

is called whenever a point was found. `x`, `y` and `z` are the coordinates of the point (where `y` is a gmp integer). `info` is the pointer that was passed to `find_points`; this can be used to store information that should persist between calls to `proc`. If `*quit` is set to a non-zero value, this indicates that `find_points` should abort the point search and return immediately; otherwise the search continues. This is how the `-1` option works. The return value is taken as a weight for counting the points; usually it will be 1.

The usual framework for using `find_points` is as follows.

```
(...)
#include "ratpoints.h"

(...)

mpz_t c[RATPOINTS_MAX_DEGREE+1]; /* The coefficients of f */
ratpoints_interval domain[2*RATPOINTS_MAX_DEGREE];
/* This contains the intervals representing the
   search region */

/*****
 * function that processes the points
 *****/

typedef struct {...} data;

int process(long x, long z, const mpz_t y, void *info0, int *quit)
{ data *info = (data *)info0;

  (...)

  return(1);
}

/*****
 * main
 *****/
```

```

int main(int argc, char *argv[])
{
    long total, n;
    ratpoints_args args;

    long degree      = 6;
    long height      = 16383;
    long sieve_primes1 = RATPOINTS_DEFAULT_SP1;
    long sieve_primes2 = RATPOINTS_DEFAULT_SP2;
    long num_primes   = RATPOINTS_DEFAULT_NUM_PRIMES;
    long max_forbidden = RATPOINTS_DEFAULT_MAX_FORBIDDEN;
    long b_low        = 1;
    long b_high        = height;
    long sturm_iter    = RATPOINTS_DEFAULT_STURM;
    long array_size    = RATPOINTS_ARRAY_SIZE;
    int no_check       = 0;
    int no_y           = 0;
    int no_reverse     = 0;
    int no_jacobi      = 0;
    int no_output      = 0;

    unsigned int flags = 0;

    data *info = malloc(sizeof(data));

    /* initialize multi-precision integer variables */
    for(n = 0; n <= degree; n++) { mpz_init(c[n]); }

    (...)

    { /* set up polynomial */
        long k;
        for(k = 0; k < 7; k++) { mpz_set_si(c[k], ...); }

        args.cof      = &c[0];
        args.degree    = 6;
        args.height    = height;
        args.domain    = &domain[0];
        args.num_inter  = 0;
        args.b_low      = b_low;
        args.b_high     = b_high;
        args.sp1        = sieve_primes1;
        args.sp2        = sieve_primes2;
        args.array_size = array_size;
        args.sturm      = sturm_iter;
        args.num_primes = num_primes;
        args.max_forbidden = max_forbidden;
        args.flags      = flags;
    }
}

```

```

info->... = ...;
(...)

total = find_points(&args, process, (void *)info);
if(total == RATPOINTS_NON_SQUAREFREE)
{ ... }
if(total == RATPOINTS_BAD_ARGS)
{ ... }

(...)

}

/* clean up multi-precision integer variables */
for(n = 0; n <= degree; n++) {mpz_clear(c[n]); }

return(0);
}

```

If points are to be searched on many curves of the same degree, then it is slightly more efficient to use the sequence

```

args.degree = degree; /* this information is needed */
find_points_init(&args);

for( ... )
{ ...
  total = find_points_work(&args, process, (void *)info);
  ...
}

find_points_clear(&args);

```

This avoids the repeated allocation and freeing of memory. Inside the loop, `cof`, `degree` and `num_inter` must be set for every curve, and so must the fields that are to be chosen afresh (see above); the others keep their values. `rptest.c` with the `-O` option does exactly this, and `make testonce` checks that the points found are the same as with every field set once per curve.

For practical examples, see `main.c` (the code that wraps `find_points` for the command line program `ratpoints`) or `rptest.c` (which runs `find_points` on some test data).

7 Fine-tuning the parameters

For large computations, it may be a good idea to try to find the best (or at least, a good) combination of parameters for the given kind of data. The default values are chosen for optimal performance on my current laptop for random genus 2 curves with small coefficients and a height bound of 100 000. For your machine and input data, other values may be better. You can replace the `testdata.h` file with your own collection of test data (and edit `rptest.c` if necessary to adapt the degree and height settings) and then time `./rptest -z` (the `-z` option suppresses the output) for various combinations of the parameter settings. `rptest` accepts most of the optional parameters of `ratpoints`, in particular the `-p`, `-N`, `-n`, `-F` and `-S` parameters, so you can easily change the

parameters used on the command line. In addition, there is an option `-h H` to change the default height bound and an option `-m m` that causes the program to repeat the computations m times.

Once you have found a good set of parameter values for your application, you can hard-code them as defaults into `ratpoints` by changing the definitions in `ratpoints.h` (and then make `distclean` all test), or you can use them to fill the `ratpoints_args` structure for your call to `find_points`.

8 Implementation

8.1 Overview

Let $F(x, z)$ be the binary form of even degree corresponding to the polynomial on the right hand side of the curve equation. The basic idea is to let run b from 1 to the height bound H , for each b , let a run from $-H$ to H , and for each coprime pair (a, b) check if $F(a, b)$ is a square.

Of course, in this form, this would take a very long time. To speed up the process, we try to eliminate quickly as many pairs (a, b) as possible before the actual test. This can be done by ‘quadratic sieving’ modulo several primes: if $F(a, b)$ is a square, it certainly has to be a square mod p , and so we can rule out all (a, b) that do not satisfy this condition. Another ingredient is to represent (for a fixed b) the various values of a by bits and treat all the bits in a ‘bit array’ (of 32 or 64 bits, as the case may be, or 128, 256 or 512 bits when SSE/AVX instructions are used) in parallel. For this, we organize the sieving information for each prime p into p arrays of p words each, one such array for every $b \bmod p$, such that the j th bit in this array is set if and only if $F(j, b)$ is a square mod p .

For each ‘denominator’ b , we then set up an array of words whose bits represent the range $-H \leq a \leq H$ (aligned so that $a = 0$ corresponds to the 0th bit of a word); the bits are initially set. Then for each of the sieving primes p , we perform a bit-wise *and* operation between this array and the sieving information at p . In a first stage, this is done on the whole array; after this first stage, each remaining (‘surviving’) non-zero word in the array is subject to tests with more primes. If some bits are still set after this second stage of sieving, the corresponding pairs (a, b) (if coprime) are then checked exactly.

In the following subsections, we discuss a number of improvements that were made.

8.2 Sorting the primes

This idea is due to **Colin Stahlke**. We do not just take the first so many primes in increasing order for the sieving, but we first compute the number of points the curve has mod p for a number of primes p and then sort the primes according to the fraction of x -coordinates that give points. We then take those primes for the sieving that have the smallest fraction of ‘surviving’ x -coordinates. In this way, we need fewer sieving primes to achieve a comparable reduction of point tests.

8.3 Using connected components

We note that we can only have points when $F(a, b)$ is non-negative. If we can determine intervals on which $f(x) = F(x, 1)$ is negative, then we do not have to look for points in these intervals. The necessary computations can be performed exactly, by computing a Sturm sequence for f and counting the number of sign changes at various points, see [Coh, Thm. 4.1.10]. This can in particular tell us whether F is negative definite, in which case we have already proved that there are no rational points in the curve. If there are real zeros, we use a subdivision method in order to find a collection of intervals containing the projections of the connected components of the curve over $\mathbb{R}$.

8.4 Using 2-adic information

John Cremona suggested that in some cases, one can determine beforehand that all points will have odd ‘numerators’ a , and so we can pack the bits more tightly by only representing odd numbers. This approach can be extended. We first find all solutions mod 16 (higher powers of 2 would be possible, but not very likely to give a significant improvement). Then for every residue class mod 16 of the denominator b , we can find the residue classes mod 16 of potential numerators a . If there are none, then we can eliminate b as a denominator altogether. If all potential a ’s are even or odd (this will always be the case for even denominators), we can restrict the sieving to such numerators.

8.5 Elimination of denominators

Depending on the equation, certain denominators can be excluded. We have seen an instance of this in the previous subsection, but we can also work with odd primes.

8.5.1 Odd degree and $\pm$ monic

If the polynomial has odd degree and leading coefficient ± 1 , then the denominator has to be a square. This reduces the time complexity tremendously.

8.5.2 Odd degree general

In general, when f has odd degree, the denominator has to be ‘almost’ a square: it must be a square times a (squarefree) divisor of the leading coefficient, and there are further restrictions on the parity of the valuation at p , for p dividing the leading coefficient, when this valuation is sufficiently large. This gives the same type of time complexity as in the monic case, but with a larger constant.

8.5.3 Even degree with non-square leading coefficient

Here we can exclude denominators divisible by a prime p such that the leading coefficient is a non-square mod p . We can also in some cases rule out denominators divisible by a certain power of p when p is a prime that divides the leading coefficient (if the leading coefficient is not a p -adic square). While computing the sieving information, we make a list of such primes and prime powers, which we then use later to eliminate denominators. In addition, we use the necessary condition that the Jacobi symbol $(\frac{l}{b'})$ must be $+1$, where l is the leading coefficient and b' is the part of b coprime to $2l$.

8.6 Reversing the polynomial

The exclusion of denominators described above is more or less effective, depending on the situation; it is the better the earlier the case was described. Since the height of the points does not change under the transformation $(x, y) \rightarrow (1/x, y/x^{\lceil n/2 \rceil})$, we can as well search on the reversed polynomial $F(z, x)$. This will result in a speed-up if the reversed polynomial belongs to a ‘better’ class than the original.

8.7 Some general remarks

Modern processors are very fast when doing basic things like moving data around between registers or simple arithmetic operations (addition, subtraction, shift, $\dots$, comparison). Even memory access can be fast when there is no cache miss. On the other hand, integer division is a slow

process. It turned out that quite some improvement of the performance was possible by removing as many instances of integer division operations as reasonably possible.

For example, we use gcd and Jacobi symbol routines that rely (almost) entirely on differences and shifts. We compute the residue classes of b modulo the various sieving primes by addition of the difference from the last b and then correcting by subtracting p a number of times if necessary, and we implement most of the testing of ‘forbidden divisors’ of b using bit arrays similar to those used for sieving the numerators.

8.8 Change log

Version 2.0 was released January 9, 2008.

Version 2.0.1 was released July 7, 2008. It fixes a bug that prevented the ‘-1’ option to work properly.

Version 2.1 was released March 9, 2009. It makes use of the SSE instructions, so that the sieve can work on 128 bits in parallel (instead of on 64 or 32). On my laptop (with an Intel Core2 processor), make test runs about 25% faster than before.

Version 2.1.1 was released April 14, 2009. It fixes a bug that in some cases prevented an early abort when *quit was set by the callback function. Thanks to **Robert Miller** for the bug report and the fix. In addition, it is now checked that `__WORDSIZE == 64` before SSE instructions are used (in `rp-private.h`), since the program then assumes that a bit array consists of two unsigned longs. In this context, `__SSE2__` has been replaced by a new macro `USE_SSE`. A further fix eliminates unnecessary copying of sieving information when SSE instructions are not used (introduced in version 2.1). This was almost always harmless, but could have resulted in memory corruption in the extreme case that all the information had to be computed for all the primes.

Version 2.1.2 was released May 27, 2009. It fixes some memory leaks. Thanks to **Robert Miller**.

Version 2.1.3 was released September 21, 2009. The library function `find_points` should now work without any bound on the degree of the polynomial. The degree bound for the `ratpoints` executable is now set to 100 by default (specified by `RATPOINTS_MAX_DEGREE`, used to be 10).

A bug in `rpctest.c` (pointed out to me by **Giovanni Mascellani** and **Randall Rathbun**) that could lead to a segmentation fault was fixed on March 10, 2011.

Version 2.2 was released January 9, 2022. The main change is that `ratpoints` can now also work with 256-bit (and, in principle, 512-bit) registers when the CPU has the relevant capabilities. Thanks to **Bill Allombert** for doing a first implementation of this. In addition, the code has been cleaned up to some extent (e.g., the variants according to register sizes are now completely dealt with through macros defined in `rp-private.h`), and some more comments have been added. There are also some further optimizations; for example, the first sieving phase now uses many of the SSE/AVX registers in parallel, and on some architectures, the code uses a faster implementation for the test whether a bit array is zero. On my current laptop, the new code (using 256-bit words) is about twice as fast as the old one (2.1.3 using 128-bit words).

Version 2.2.1 was released January 18, 2022. This fixes a small bug that was introduced in the previous version, which could lead to the program missing points at the upper end of the search interval for the numerators. Thanks to **Bill Allombert**.

Version 2.2.2 was released May 15, 2023. This fixes a small bug that led to the polynomials being reversed incorrectly when the degree is odd. Thanks to **Nicolas Mascot** and **Bill Allombert**.

Version 2.2.3 was released September 6, 2026. It fixes the code path using 512-bit registers, which had never been exercised. Two problems were found. The first one would have made the program

crash: the array `sieves0` of precomputed sieving information is declared as an array of unsigned long, but it is read through pointers to `ratpoints.bit_array`, and gcc aligns such a global array to at most 32 bytes, whereas a 512-bit bit-array needs 64. The aligned 512-bit load that gcc generates in the second sieving phase would then fault; whether it did so depended on where the linker happened to place the array. The array is now given the alignment of `ratpoints.bit_array` explicitly, in every variant. The second problem was that the `TEST` macro, which is used in the innermost loop of the second phase, still used the generic fall-back version; gcc compiles this to a spill of the whole 64-byte vector to the stack followed by reading it back word by word, which defeats store-to-load forwarding. It now uses `vpptestmq`, in analogy with the 256-bit version.

Since I have no CPU with AVX512F available, this was tested by compiling with `-DUSE_AVX512` but without `-mavx512f`, so that gcc expresses the 64-byte vectors through 256-bit operations. This exercises the whole code path (in particular `RBA_PACK == 8` and the two mask macros); `make test1` and `make test2` both reproduce the reference output exactly, as do 400 randomly generated comparisons against the 256-bit version. What this cannot check is the genuine 512-bit instructions. In the meantime, **Drew Sutherland** has run the two variants against each other on an AMD Ryzen 9 9950X3D (Zen 5 architecture), for a curve with many rational points and a height bound of 10^6 ; the 512-bit version was faster by 13 to 14%.

Version 2.2.4 was released September 13, 2026. It fixes bugs that a review of the code turned up. A polynomial of degree 1 — given as such, or reached from $y^2 = c_2x^2 + c_1x$ by the reversal, or from a zero leading coefficient — made the Sturm sequence code read past the end of its arrays and crash; this affects library users as well. When the positivity region of f did not meet the search domain, the program returned before it had looked at the points at infinity, so that `ratpoints -10000000000 0 0 1' 45` printed nothing where `-s` printed `(1 : 0 : 0)`, and after a reversal the lost point is an affine point of the given curve. When no residue class of the denominator modulo 16 admitted any numerator, the array of numerator patterns was read before it had been filled; by default the answer was still right, but with `-j -F 0` the program printed nondeterministic points outside the height bound after a run several thousand times longer than necessary. And several input fields of `ratpoints_args` were used as scratch space — the number of forbidden divisors found was written into `max_forbidden`, the intersected search intervals into `num_inter`, normalised values into `sp1`, `sp2` and others — so that a program which fills the structure once and then calls `find_points_work` for curve after curve, as the section on the library suggests, had the forbidden-divisor test switched off after the first curve without forbidden divisors (a factor of two in running time) and the first curve's positivity region intersected into the search domains of the others, which loses points. The program no longer writes either, and the manual now says exactly which fields it does set; `make test1once` runs the test curves with the structure filled once, and `make test3` runs one or more invocations of `ratpoints` per bug against a reference. Also fixed: four left shifts of negative values (undefined in C, harmless with gcc so far), two arrays that could be declared with length zero under `-n 0 -N 0`, an overflow in the enumeration of the divisors of the leading coefficient for denominator bounds above about $4 \cdot 10^{15}$, and two constants in `ratpoints.h` that could not be overridden from the compiler command line. The library is binary compatible with 2.2.3.

References

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